

# DATABASE DESIGN

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| Route_points |         |
|--------------|---------|
| point_id     | integer |
| route_id     | integer |
| latitude     | integer |
| longitude    | integer |
| point_order  | integer |

| Collaborators   |         |
|-----------------|---------|
| collaborator_id | integer |
| route_id        | integer |
| user_id         | integer |

| Routes       |           |
|--------------|-----------|
| is_public    | boolean   |
| created_at   | timestamp |
| route_id     | integer   |
| user_id      | integer   |
| route_name   | varchar   |
| last_updated | timestamp |

| TipComments  |           |
|--------------|-----------|
| comment_id   | integer   |
| tip_id       | integer   |
| comment_text | text      |
| created_at   | timestamp |
| user_id      | integer   |

| Tips       |           |
|------------|-----------|
| tip_id     | integer   |
| tip_text   | text      |
| created_at | timestamp |
| user_id    | integer   |

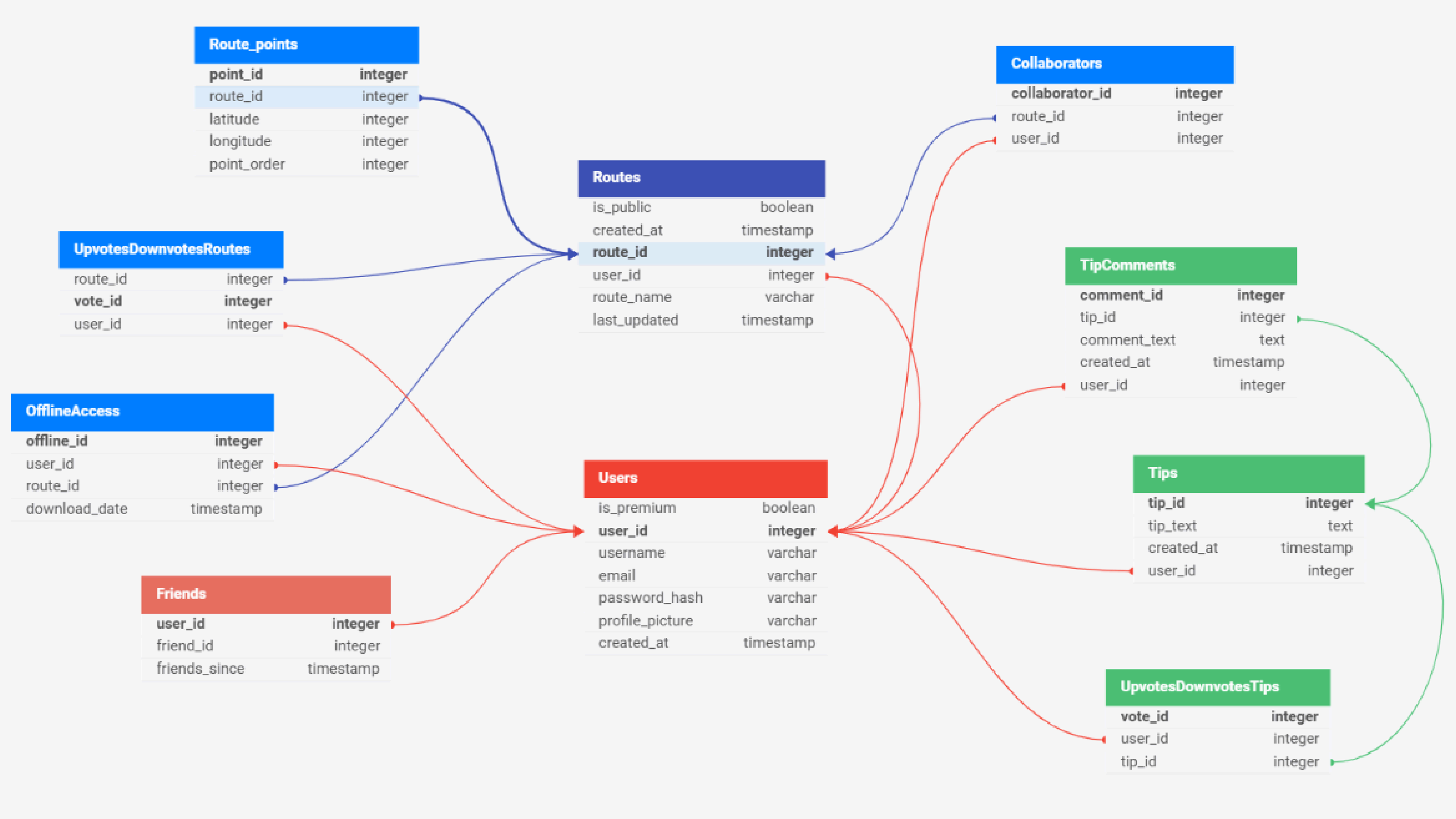
| UpvotesDownvotesTips |         |
|----------------------|---------|
| vote_id              | integer |
| user_id              | integer |
| tip_id               | integer |

| UpvotesDownvotesRoutes |         |
|------------------------|---------|
| route_id               | integer |
| vote_id                | integer |
| user_id                | integer |

| OfflineAccess |           |
|---------------|-----------|
| offline_id    | integer   |
| user_id       | integer   |
| route_id      | integer   |
| download_date | timestamp |

| Friends       |           |
|---------------|-----------|
| user_id       | integer   |
| friend_id     | integer   |
| friends_since | timestamp |

| Users           |           |
|-----------------|-----------|
| is_premium      | boolean   |
| user_id         | integer   |
| username        | varchar   |
| email           | varchar   |
| password_hash   | varchar   |
| profile_picture | varchar   |
| created_at      | timestamp |



# TABLE RELATIONSHIPS AND FLOW

- **Users** own routes and can share them with friends or collaborators.
- **Routes** can have multiple waypoints (**RoutePoints**) to define the trip path.
- **Collaborators** table supports multi-user route editing.
- **UpvotesDownvotesRoutes** table allows route rating.
- Users can add **tips** to enrich the travel experience with additional information and build a community.
- **UpvotesDownvotesTips** and **TipsComment** tables allow users to rank tips and comment on them.
- **Friends** table facilitates route sharing and collaboration.
- Routes are downloadable through the **OfflineAccess** table for offline use.

# RELATIONSHIP TYPES

- **UpvotesDownvotesRoutes** to **Routes** - one to many; each user can rate many routes, but each rating is assigned to only one user
- **Routes** to **Users** - one to many; each user can create many routes, but each route has only one creator
- **UpvotesDownvotesTips** to **Users** - one to many; each user can rate many tips, but each rating is assigned to only one user
- **Tips** to **Users** - one to many; each user can create many tips, but each tip has only one creator
- **RoutePoints** to **Routes** - many to many; each route can have many points and one point can be assigned to many routes
- **Collaborators** to **Routes** - many to many; each route can have a few collaborators and one collaborator can be assigned to many routes
- **TipComments** to **Comments** - one to many; each tip can have many comments, but each comment is assigned to one route

# RELATIONSHIP TYPES

- **Friends to Users** - one to many; each user can have many friends, but each friend is assigned to only one user
- **OfflineAccess to Users** - one to many; each user can have many
- **UpvotesDownvotesRoutes to Users** - one to many; each user can rate many routes, but each rating is assigned to only one user
- **OfflineAccess to Routes** - one to many; each route can be downloaded multiple times, but each offline access is assigned to one route
- **UpvotesDownvotesTips to Tips** - one to many; each tip can have many ratings, but each rating is assigned to only one tip
- **Collaborators to Users** - one to many; each users can be many collaborators, but each collaborator is one user
- **TipsComments to Users** - one to many; each users can have many comments, but each comment is assigned to one user

THANK YOU  
FOR YOUR  
ATTENTION

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